Backend Assignment - Transporter Assignment on Lanes

You are tasked with designing and implementing a **Spring Boot application** to solve the "Transporter Assignment on Lanes" problem. The application must optimize the assignment of transporters to lanes across India, ensuring the following objectives are met:

1. **Cost Minimization**: Minimize the total cost of assigning transporters to lanes.
2. **Maximize Transporter Usage**: Utilize the maximum number of transporters up to a user-specified limit.
3. **Full Lane Coverage**: Ensure every lane is assigned to at least one transporter.

The application must expose **two APIs**:

1. **API for Inputting Data**: Allows users to input data for lanes and transporter quotes.
2. **API for Fetching Results**: Provides optimized transporter assignments based on the constraints and the specified maximum number of transporters.

API Specifications

API 1: Submit Input Data

Request URL:

```
POST /api/v1/transporters/input
```

Request Body:

```
{
  "lanes": [
    {"id": 1, "origin": "Mumbai", "destination": "Delhi"},
    {"id": 2, "origin": "Delhi", "destination": "Bangalore"},
    {"id": 3, "origin": "Chennai", "destination": "Kolkata"},
    {"id": 4, "origin": "Pune", "destination": "Hyderabad"},
    {"id": 5, "origin": "Ahmedabad", "destination": "Jaipur"},
    {"id": 6, "origin": "Guwahati", "destination": "Shillong"},
    {"id": 7, "origin": "Lucknow", "destination": "Patna"},
    {"id": 8, "origin": "Ranchi", "destination": "Bhubaneswar"},
    {"id": 9, "origin": "Surat", "destination": "Indore"},
    {"id": 10, "origin": "Nagpur", "destination": "Raipur"}
  ],
```

```
  "transporters": [
    {"id": 1, "name": "Transporter T1", "laneQuotes": [{"laneId": 1,
"quote": 5000}, {"laneId": 2, "quote": 6000}]},
    {"id": 2, "name": "Transporter T2", "laneQuotes": [{"laneId": 3,
"quote": 7000}, {"laneId": 4, "quote": 4000}]},
    {"id": 3, "name": "Transporter T3", "laneQuotes": [{"laneId": 5,
"quote": 3000}, {"laneId": 6, "quote": 2000}]},
    {"id": 4, "name": "Transporter T4", "laneQuotes": [{"laneId": 7,
"quote": 8000}, {"laneId": 8, "quote": 5000}]},
    {"id": 5, "name": "Transporter T5", "laneQuotes": [{"laneId": 9,
"quote": 4500}, {"laneId": 10, "quote": 3500}]},
    {"id": 6, "name": "Transporter T6", "laneQuotes": [{"laneId": 1,
"quote": 5500}, {"laneId": 2, "quote": 6200}]},
    {"id": 7, "name": "Transporter T7", "laneQuotes": [{"laneId": 3,
"quote": 7500}, {"laneId": 4, "quote": 4500}]}
  ]
}
```


Response:


```
{
  "status": "success",
  "message": "Input data saved successfully."
}
```

API 2: Get Optimized Results

Request URL:

```
POST /api/v1/transporters/assignment
```

Request Body:


```
{
  "maxTransporters": 3
}
```


Response (Example):


```
{
```

```
  "status": "success",
  "totalCost": 29000,
  "assignments": [
    {"laneId": 1, "transporterId": 1},
    {"laneId": 2, "transporterId": 1},
    {"laneId": 3, "transporterId": 2},
    {"laneId": 4, "transporterId": 2},
    {"laneId": 5, "transporterId": 3},
    {"laneId": 6, "transporterId": 3},
    {"laneId": 7, "transporterId": 2},
    {"laneId": 8, "transporterId": 3},
    {"laneId": 9, "transporterId": 1},
    {"laneId": 10, "transporterId": 1}
  ],
  "selectedTransporters": [1, 2, 3]
}
```

Sample Test Input Data

Test Case 1

API 1: Submit Input Data
Drop Box folder Link - https://www.dropbox.com/scl/fo/fevep9iw5gcog1dixb4lr/AGrKYB-EwJEtYBVPn5ir5Fg?rlkey=vgmau9ntgl4w9nh47o9b8z035&st=qt5dpuqk&dl=0

Request Body:

```
{

  "lanes": [

    {"id": 1, "origin": "Mumbai", "destination": "Delhi"},

    {"id": 2, "origin": "Delhi", "destination": "Bangalore"},

    {"id": 3, "origin": "Chennai", "destination": "Kolkata"},

    {"id": 4, "origin": "Pune", "destination": "Hyderabad"},

    {"id": 5, "origin": "Ahmedabad", "destination": "Jaipur"}

  ],
```

```
  "transporters": [

    {"id": 1, "name": "Transporter T1", "laneQuotes": [{"laneId": 1,
"quote": 20835}, {"laneId": 2, "quote": 10512}, {"laneId": 3,
"quote": 22105}, {"laneId": 4, "quote": 42481}, {"laneId": 5,
"quote": 19862}]},

    {"id": 2, "name": "Transporter T2", "laneQuotes": [{"laneId": 1,
"quote": 48844}, {"laneId": 2, "quote": 31326}, {"laneId": 3,
"quote": 18640}, {"laneId": 4, "quote": 45828}, {"laneId": 5,
"quote": 18297}]},

    {"id": 3, "name": "Transporter T3", "laneQuotes": [{"laneId": 1,
"quote": 39020}, {"laneId": 2, "quote": 20648}, {"laneId": 3,
"quote": 31438}, {"laneId": 4, "quote": 36447}, {"laneId": 5,
"quote": 12789}]},

    {"id": 4, "name": "Transporter T4", "laneQuotes": [{"laneId": 1,
"quote": 14400}, {"laneId": 2, "quote": 44514}, {"laneId": 3,
"quote": 14316}, {"laneId": 4, "quote": 10678}, {"laneId": 5,
"quote": 13032}]},

    {"id": 5, "name": "Transporter T5", "laneQuotes": [{"laneId": 1,
"quote": 11601}, {"laneId": 2, "quote": 19760}, {"laneId": 3,
"quote": 40870}, {"laneId": 4, "quote": 20635}, {"laneId": 5,
"quote": 26421}]},

    {"id": 6, "name": "Transporter T6", "laneQuotes": [{"laneId": 1,
"quote": 35095}, {"laneId": 2, "quote": 12494}, {"laneId": 3,
"quote": 17808}, {"laneId": 4, "quote": 36210}, {"laneId": 5,
"quote": 39444}]},

    {"id": 7, "name": "Transporter T7", "laneQuotes": [{"laneId": 1,
"quote": 26070}, {"laneId": 2, "quote": 41098}, {"laneId": 3,
"quote": 20932}, {"laneId": 4, "quote": 16897}, {"laneId": 5,
"quote": 27938}]}

  ]

}
```

API 2: Get Optimized Results

Request Body:

```
{
  "maxTransporters": 3
}
```

Response (Example):

```
{
  "status": "success",
  "totalCost": 75000,
  "assignments": [
    {"laneId": 1, "transporterId": 4},
    {"laneId": 2, "transporterId": 6},
    {"laneId": 3, "transporterId": 2},
    {"laneId": 4, "transporterId": 4},
    {"laneId": 5, "transporterId": 3}
  ],
  "selectedTransporters": [4, 6, 3]
}
```

This test case ensures that the application adheres to the constraints while minimizing costs and utilizing up to three transporters.

Test Case 2

Lanes:

```
[
  {"id": 1, "origin": "Chandigarh", "destination": "Shimla"},
  {"id": 2, "origin": "Agra", "destination": "Kanpur"},
  {"id": 3, "origin": "Varanasi", "destination": "Gorakhpur"},
  {"id": 4, "origin": "Amritsar", "destination": "Ludhiana"},
```

```
  {"id": 5, "origin": "Coimbatore", "destination": "Madurai"}
]
```


Transporters:


```
[
  {"id": 1, "name": "Transporter X", "laneQuotes": [{"laneId": 1,
"quote": 1000}, {"laneId": 2, "quote": 1500}]},
  {"id": 2, "name": "Transporter Y", "laneQuotes": [{"laneId": 3,
"quote": 2000}, {"laneId": 4, "quote": 1800}]},
  {"id": 3, "name": "Transporter Z", "laneQuotes": [{"laneId": 5,
"quote": 1200}]},
  {"id": 4, "name": "Transporter W", "laneQuotes": [{"laneId": 1,
"quote": 1100}, {"laneId": 4, "quote": 1750}]}
]
```
